

456 Project—Implicit and Explicit Errors on a 1-D Heat Equation

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Introduction

The purpose of this project is to analyze the differences in the implicit and explicit Euler methods when solving a one-dimensional heat equation. Using the Method of Lines, we discretize the spatial dimension, turning the PDE into a system of ODEs at each discrete point in space.

Specifically, we will investigate the effect of the stability parameter $r = \frac{k\Delta t}{(\Delta x)^2}$ on the global error and stability of each method. The heat equation we consider is:

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}, \quad k = \frac{1}{25}, \quad 0 \leq x \leq 1, \quad 0 \leq t \leq 1$$

Subject to Nuemann boundary conditions and the initial temperature distribution:

$$\frac{\partial u}{\partial x}(0, t) = 0, \quad \frac{\partial u}{\partial x}(1, t) = 0, \quad u(x, 0) = 5 + 4\cos(3\pi x)$$

With the analytic solution:

$$u(x, t) = 5 + 4\cos(3\pi x)e^{-\frac{9}{25}\pi^2 t}$$

As the Method of Lines creates ODEs respective to time, we will vary Δt while Δx remains constant to sample our different values of r .

Numerical Methodology

To implement the Method of Lines we first define a spatial grid: $x_i = i\Delta x$ so that we can approximate $\frac{\partial^2 u}{\partial x^2}$ across it. To do this we implement the Second-Order Difference Formula:

$$\frac{\partial^2 u}{\partial x^2} \approx \frac{u_{i-1} - 2u_i + u_{i+1}}{(\Delta x)^2}$$

This is where we must address our trivial Neumann boundary conditions, using ghost nodes. To do this, when we approximate our Laplacian, we set the boundary nodes to be equal to artificial nodes just outside our bounds so that $u_{-1} = u_1$ and $u_{N-1} = u_{N+1}$. Substituting this into the equation above we obtain

$$\frac{\partial^2 u_0}{\partial x^2} \approx 2 \frac{u_1 - u_0}{(\Delta x)^2}, \quad \frac{\partial^2 u_N}{\partial x^2} \approx 2 \frac{u_{N-1} - u_N}{(\Delta x)^2}$$

Taking this and plugging it into our heat equation yields:

$$\frac{du_i}{dt} = \frac{k}{(\Delta x)^2}(u_{i-1} - 2u_i + u_{i+1})$$

Thus, to formulate our explicit method we take:

$$\begin{aligned} u_i^{n+1} &= u_i^n + \Delta t f(u_i^n) \\ &= u_i^n + \frac{k\Delta t}{(\Delta x)^2}(u_{i-1}^n - 2u_i^n + u_{i+1}^n) \end{aligned}$$

And for our implicit method:

$$\begin{aligned} u_i^{n+1} &= u_i^n + \Delta t f(u_i^{n+1}) \\ &= u_i^n + \frac{k\Delta t}{(\Delta x)^2}(u_{i-1}^{n+1} - 2u_i^{n+1} + u_{i+1}^{n+1}) \end{aligned}$$

Now, after substituting $r = \frac{k\Delta t}{\Delta x^2}$, we rearrange into the form:

$$-ru_{i-1}^{n+1} + (1 + 2r)u_i^{n+1} - ru_{i+1}^{n+1} = u_i^n$$

Then, as this holds for all i in our spatial grid, this becomes a tridiagonal system of the form:

$$\begin{bmatrix} 1+2r & -2r & 0 & \dots & 0 \\ -r & 1+2r & -r & \ddots & \vdots \\ 0 & \ddots & \ddots & \ddots & 0 \\ \vdots & \ddots & -r & 1+2r & -r \\ 0 & \dots & 0 & -2r & 1+2r \end{bmatrix} \begin{bmatrix} u_0^{n+1} \\ u_1^{n+1} \\ \vdots \\ u_{N-1}^{n+1} \\ u_N^{n+1} \end{bmatrix} = \begin{bmatrix} u_0^n \\ u_1^n \\ \vdots \\ u_{N-1}^n \\ u_N^n \end{bmatrix}$$

Where the first and last rows embed the trivial Nuemann boundaries conditions. To obtain u_i^{N+1} from this, we will implement a linear solve program.

From this we see the motivation for varying Δt while keeping Δx constant, as it is the step size we used to derive both of our methods.

To evaluate our error, we will use the L_2 and L_∞ norms of the difference in our numerical and analytical solutions. We take this difference at $t = 1$, allowing any numerical instabilities or truncation errors sufficient time to propagate, giving us an accurate com-

parison of model performance.

$$L_2 = \sqrt{\sum_{i=0}^N e_i^2}, \quad L_\infty = \max_i |e_i|, \quad e_i = u_{num}(x_i, 1) - (5 + 4\cos(3\pi x_i)e^{-\frac{9}{25}\pi^2})$$

Stability Expectations

To determine the stability restriction on the explicit Euler method applied to the heat equation, we apply a von Neumann (Fourier) stability analysis. Thus, we assume a solution of the form:

$$u_i^n = G^n e^{i\theta i},$$

where $\theta \in [0, \pi]$ is the wave number and G is the amplification factor. Substituting this ansatz into our update formula derived above

$$u_i^{n+1} = u_i^n + r(u_{i-1}^n - 2u_i^n + u_{i+1}^n), \quad r = \frac{k\Delta t}{\Delta x^2},$$

yields

$$Ge^{i\theta i} = e^{i\theta i} [1 + r(e^{-i\theta} - 2 + e^{i\theta})].$$

Removing the common factor $e^{i\theta i}$, we obtain

$$G = 1 + r(e^{-i\theta} - 2 + e^{i\theta}).$$

Using the identity $e^{i\theta} + e^{-i\theta} = 2\cos\theta$, this becomes

$$G = 1 + 2r(\cos\theta - 1).$$

Applying the half angle identity

$$\cos\theta - 1 = -2\sin^2\left(\frac{\theta}{2}\right),$$

we obtain the real amplification factor

$$G(\theta) = 1 - 4r\sin^2\left(\frac{\theta}{2}\right).$$

For stability, we require that all harmonic modes satisfy

$$|G(\theta)| \leq 1 \quad \text{for all } \theta.$$

Then, as $\sin^2(\theta/2) \in [0, 1]$, the minimum value of G occurs at $\sin^2(\theta/2) = 1$, and therefore, $G_{\min} = 1 - 4r$. The condition $|G_{\min}| \leq 1$ yields

$$1 - 4r \geq -1 \quad \implies \quad 4r \leq 2 \quad \implies \quad r \leq \frac{1}{2}.$$

Thus, the explicit Euler method is stable if and only if

$$0 \leq r \leq \frac{1}{2}.$$

This is the standard CFL condition for the heat equation, illuminating why we expect our explicit method to diverge significantly from our exact solution for larger values of r .

For our implicit method, we recognize that our update formula

$$u_i^{n+1} = u_i^n + r(u_{i-1}^{n+1} - 2u_i^{n+1} + u_{i+1}^{n+1})$$

yields the real amplification factor

$$G(\theta) = \frac{1}{1 + 4r \sin^2\left(\frac{\theta}{2}\right)}.$$

Where it is clear that for $r \geq 0$, all values of θ satisfy our inequality

$$|G(\theta)| \leq 1.$$

Because of this, our implicit method is said to be unconditionally stable, and we expect to see numerical stability in all regions, even those where our explicit method becomes unstable.

Results and Analysis

Our numerical experiments demonstrate the stark contrast between the behaviors of our implicit and explicit Euler methods as the stability condition, r , increases.

As derived in the stability analysis, our explicit method is subject to the bound $r \leq \frac{1}{2}$.

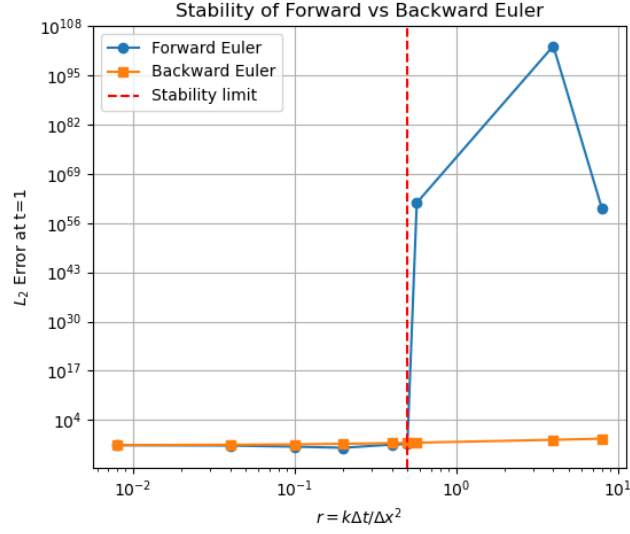


Figure 1: L_2 Error Against Step Size r

This condition is empirically validated by our results, as seen in Figure 4, a logarithmic graph of the L_2 error norm at $t=1$ for different step sizes in our stability condition. For $r < \frac{1}{2}$ the error of the explicit method remains stable and comparable to that of the implicit method's. Immediately after the condition is violated, we see an explosive growth in the error, reaching magnitudes $> 10^{100}$. Meanwhile, the error of our implicit method remains flat across all values of r , clearly displaying the unconditional nature of its stability.

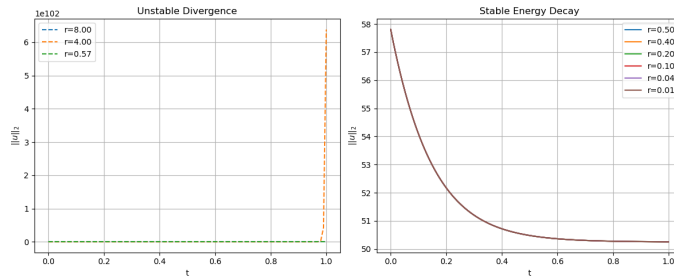


Figure 2: Forward Modes: Unstable Divergence and Stable Energy Dissipation

The consequence of breaking our stability condition is well described by the time evolution of the 'energy' of our heat system $\|u\|_2$. In the heat equation with $u(x, 0) =$

$5 + 4\cos(3\pi x)$, we expect to see dissipation of this energy over time, corresponding to decaying $\|u\|_2$. Figure 3 illustrates that for values of r satisfying the stability condition this character is displayed, while those that exceed it display non-physical unstable divergence characterized by growing errors from an amplification factor greater than one.

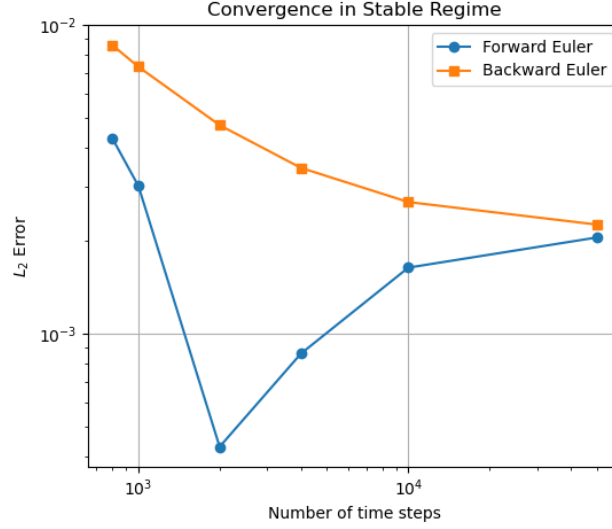


Figure 3: Stable Regime Convergence Trends

In the region where both methods are stable, we can investigate their convergence properties more closely. Figure 5 shows that under proper conditions, our explicit method is even more accurate than our implicit, though, for very small step sizes, error does begin to increase. This suggests that while the implicit method offers strong stability, it does not necessarily offer improved accuracy to the explicit Euler within its region of stability.

Conclusion

This project was successful in verifying the stability criteria for the explicit method and the unconditional stability of the explicit method on our 1-D heat equation. Our numerical implementation of the explicit method displayed a clear change in behavior from accurate decay within $r \leq \frac{1}{2}$ to non-physical exponential growth above the bound. In contrast, our implicit method showed constant convergence across all regimes of r ,

producing physical results even for large time steps.

While the implicit method is more computationally taxing due to the linear solve of the tridiagonal system, for a heat equation that is more stiff (greater k) or with a high-resolution spatial grid it is superior, as the stability condition of the explicit method becomes very limiting. Possible extensions to this project include increasing the spatial dimensions or implementing other PDEs using the Method of Lines.

References

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- Smith, G. D. (1985). Numerical Solution of Partial Differential Equations: Finite Difference Methods. Oxford University Press.